
Quiz 1- Make Up

Important Notes: By submitting this exam, you affirm that you have not given or received any unauthorized assistance during this test, and that all work is your own. Academic integrity is highly valued in this course, and I am committed to ensuring a fair testing environment for everyone. Cheating undermines trust and devalues the hard work of others. Be advised that any form of cheating will not be tolerated and will result in disciplinary action, up to and including failure of the course.

Exam Guidelines

- This exam is closed book and notes and **the use of the internet is strictly prohibited!** However, you are free to use a hand-written, double-sided, A4 size “cheat” sheet with your notes.
- **Communication among students is not allowed** during the exam. This includes talking, text messaging, emailing, chatting, forum discussions, etc. Headphones are also not permitted.
- All questions, irrespective of their nature, should be directed only to Prof. Kaya or the teaching assistants.

Submission Guidelines and Deadlines

- When the exam concludes, all papers and the handwritten cheat sheet must be submitted to the TAs or Dr. Kaya.
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AgriCoop is preparing its seasonal production plan for two crops, Wheat and Barley. Each acre of land planted produces one unit of crop that can be sold at the market price of \$480 (Wheat) or \$520 (Barley). Buying one unit of fertilizer costs the cooperative \$25 each, and irrigation hours cost \$15 each.

Planting one acre of Wheat requires \$200 in baseline expenses, 5 units of fertilizer, and 3 irrigation hours. Planting one acre of Barley requires \$220 in baseline expenses, 4 units of fertilizer, and 5 irrigation hours.

This season AgriCoop has 900 units of fertilizer in stock and 1,200 irrigation hours available from its water system.

Management wants to determine how many acres of Wheat and Barley to cultivate in order to maximize profit, given revenues, expenses, and limited resources.

- (a) [20 points] Formulate a linear programming model to determine the production plan to maximize total profit using only the resources in stock. Clearly define the decision variables, objective function, and all constraints.
- (b) [20 points] Graph the feasible region of the problem in the two-dimensional plane. Clearly label all constraints and indicate the feasible set/region. Solve the problem graphically and determine:
 - the optimal production quantities of Wheat and Barley,
 - the maximum achievable profit.
- (c) [15 points] At the optimal solution, identify which constraints are binding and which are non-binding. Briefly explain what this means in practical terms (in real life).
- (d) [25 points] Find the dual of this problem. Interpret the shadow price associated with each constraint: fertilizer and irrigation hours. Explain what each value represents economically.
- (e) [20 points] Suppose AgriCoop had a set of crops P and a set of resources R , with each crop $p \in P$ yielding revenue r_p , requiring a_{rp} units of each resource $r \in R$. Also, each resource r is available in capacity b_r . Write the general linear program formulation in closed algebraic form using sets, indices, and parameters.

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Solution

(a) Linear Programming Formulation (20 points)

Decision variables:

x_W = acres of **Wheat** to plant (one acre $\Rightarrow$ one unit sold).

x_B = acres of **Barley** to plant (one acre $\Rightarrow$ one unit sold).

Objective Function:

Per-unit profit (revenue – baseline expenses – resource cash costs):

$$\pi_W = 480 - 200 - 5(25) - 3(15) = 110, \quad \pi_B = 520 - 220 - 4(25) - 5(15) = 125.$$

Primal LP:

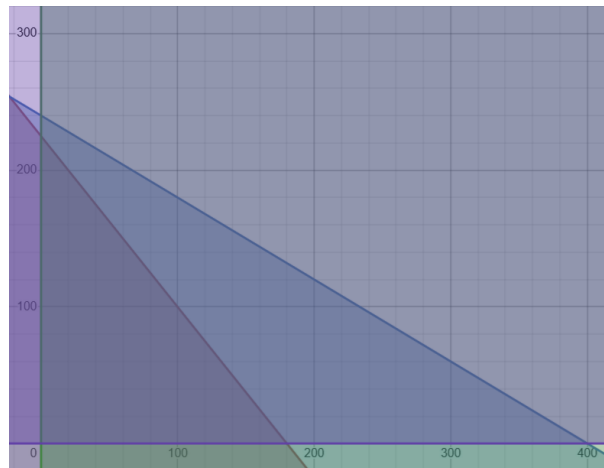
$$\begin{aligned} \max \quad & 110x_W + 125x_B \\ \text{s.t.} \quad & 5x_W + 4x_B \leq 900 && (\text{fertilizer units}) \\ & 3x_W + 5x_B \leq 1200 && (\text{irrigation hours}) \\ & x_W, x_B \geq 0. \end{aligned}$$

- Defining the decision variables in a clear way: 5
 - If they do not write down that it is the **acres** of wheat/barley to be planted: -1
- Defining the objective function in a clear way: 5
 - If they do not explain it verbally: -1
 - If there is a mistake in the coefficients: -1 (be careful this mistake may carry for the rest of the solution!)
- Raw material Non-negativity constraints: 10
 - If they do not explain the constraint verbally (except the non negativity): -0.5 for each constraint
 - If there is a mistake in the coefficients: -0.5 for each constraint (be careful this mistake may carry for the rest of the solution!)

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(b) Graphical Solution (20 points)

The feasible region is defined by the following constraints and in two-dimensional plane should look like the graph below:



Corner points:

$$(0,0), \quad (180,0) \text{ from } 5x_W \leq 900, \quad (0,225) \text{ from } 4x_B \leq 900.$$

(The intersection of the two resource lines is infeasible: it yields $x_W < 0$.)

Objective values:

$$z(180,0) = 110 \cdot 180 = 19,800, \quad z(0,225) = 125 \cdot 225 = \boxed{28,125}.$$

Thus the **optimal plan** is $x_W^* = 0, x_B^* = 225$ with **maximum profit** $z^* = 28,125$.

- Correct graph and labeled constraints: 8
 - If there is a major problem with the graph but they have attempted solving: -6 (be careful this mistake may carry for the rest of the solution!)
 - If they did not label the constraints: -1
- Feasible region identified: 2
- Corner point evaluation: 6
 - If they could not correctly identify all 3 corner points, but did 2 of them: -2
- Optimal solution and profit: 4

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(c) Binding and Non-binding Constraints (15 points)

At the optimal solution:

Binding constraints at (0, 225):

$$5(0) + 4(225) = 900 \text{ (fertilizer binds),} \quad 3(0) + 5(225) = 1125 < 1200 \text{ (irrigation slack 75).}$$

Interpretation: Only the fertilizer constraint is binding. Additional fertilizer would allow higher profits.

- Binding constraints identified: 7
- Non-binding constraints identified: 4
- Interpretation: 4

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(d) Dual Problem and Shadow Prices (25 points)

Let y_1 = shadow price of one unit of fertilizer; y_2 = shadow price of one irrigation hour.

$$\begin{aligned} \min \quad & 900y_1 + 1200y_2 \\ \text{s.t.} \quad & 5y_1 + 3y_2 \geq 110 \quad (\text{covers Wheat's unit profit}) \\ & 4y_1 + 5y_2 \geq 125 \quad (\text{covers Barley's unit profit}) \\ & y_1, y_2 \geq 0. \end{aligned}$$

Interpretation: y_1 is the marginal increase in optimal profit per extra unit of fertilizer; y_2 per extra irrigation hour.

Dual solution (from CS): $x_B^* > 0 \Rightarrow 4y_1 + 5y_2 = 125$; irrigation has slack $\Rightarrow y_2^* = 0$. Hence

$$\boxed{y_1^* = 31.25, \quad y_2^* = 0}.$$

Strong duality check: $900(31.25) + 1200(0) = \boxed{28,125} = z^*$.

Interpretation:

- One additional unit of fertilizer increases maximum profit by \$31,25.
- One additional unit of irrigation hours increases maximum profit by \$0.
- Correct dual formulation: 10
 - Minor coefficient mistake: -1
 - Major mistake: - 5
- Correct shadow prices: 8
 - Could not use complementary slackness to make constraints tight: -4
- Clear economic interpretation: 7
 - Included an explanation, but it's not clear: -2
 - The interpretation is completely wrong: -5

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(e) General Linear Program Formulation (20 points)

Let P be the set of crops, R the set of resources.

Parameters:

r_p = profit per unit (per acre) of crop p ,

a_{rp} = units of resource r per unit p ,

b_r = available units of resource r .

Decision variables:

$x_p \geq 0$ (acres/units of crop p).

General primal:

$$\begin{aligned} \max \quad & \sum_{p \in P} r_p x_p \\ \text{s.t.} \quad & \sum_{p \in P} a_{rp} x_p \leq b_r \quad \forall r \in R, \\ & x_p \geq 0 \quad \forall p \in P. \end{aligned}$$

- Sets, parameters, and indices are defined: 6
- Objective function: 6
- Constraints: 6
- Clarity and notation: 2